Terms of Reference

Working Group 1: Global Cryosphere Monitoring

The Decade of Action for Cryospheric Sciences (2025-2034)

Background

On August 13, 2024, the United Nations General Assembly adopted Resolution 78/321, proclaiming 2025 to 2034 as the Decade of Action for Cryospheric Sciences. The resolution aims to accelerate efforts to address the urgent challenges posed by the melting cryosphere through enhanced scientific research, monitoring, and international cooperation. The resolution highlights the need to understand the cryosphere's role in the Earth's climate system, freshwater resources, ecosystems, and human social, political, and economic activities. It also calls for raising awareness of cryospheric loss and encourages the sharing of best practices to build resilience against related risks. Member States, UN bodies, and stakeholders are invited to support these activities. UNESCO is tasked with leading implementation of the Decade, with other UN organizations and partners, using voluntary contributions to develop programmes.

During the first year of the Decade, 2025, an Ad Hoc Committee was formed by UNESCO to form the structures necessary to implement the Decade. These are summarised in a **Synthesis Document**, which outlines the vision for the Decade. As a result of this process, the committee identified five Overarching Challenges (OC) for the Decade:

OC1. Global Cryosphere Monitoring
OC2 Actionable Cryosphere Projections
OC3 Cryospheric Risk Management Solutions
OC4 Adaptation to Cryosphere Loss
OC5 Amplify the Impact of Cryospheric Science

Within the Decade, Working Groups aligned with each OC serve as the *main engines of delivery*. The role of Working Group 1 is to achieve the Main Goals (MG) of **OC1, Global Cryosphere Monitoring**:

MG1.1: Build a Global Cryosphere Monitoring Inventory
MG1.2: Enhance Monitoring Capabilities
MG1.3: Make Observations Actionable

The Decade is split into three triennia. Each MG is assigned a 3-year block, marked by a triennial summit. It is anticipated that appointment to each working group will be on an initial three-year basis, with membership rotating as focus shifts towards subsequent MGs.

By the end of the Decade, this Working Group should achieve a global cryosphere observation framework as critical research infrastructure through coordination of existing and emerging observation systems, that maximises the value of existing and future investments, and provides the observational framework for modelling, hazard assessment, adaption planning and informed decision-making.

Initial Objectives

Based on the Decade Synthesis document, Working Group 1 is tasked with delivering progress towards OC1, Global Cryosphere Monitoring. The focus of the *first three years is on MG1.1: Build a Global Cryosphere Monitoring Inventory*. The working group should build a metadata inventory (catalogue) of existing cryosphere observations, strengthening coordination across current efforts. This inventory will act as a meta-database that provides data visibility across different sectors. The working group will then identify and assess critical monitoring gaps which initially considers:

- Spatial and elevational gaps, particularly in high-mountain, cold regions and circumpolar regions.
- Variable gaps, where key parameters such as snow water equivalent, permafrost presence, soil temperature or glacier mass balance are undermeasured.
- Methodological gaps, where technological advances could enhance monitoring capabilities.

Towards the end of the first three-year period (2026-2029), focus should move towards MG1.2: Enhance Monitoring Capabilities.

The work of Working Group 1 over the first three years will culminate in a report to the first *Triennial Summit of the Decade of Action on Cryosphere Sciences*.

Wider Mandate and Deliverables

As one of the Decade's five thematic Working Groups, the Working Group 1 is mandated to:

- Act as the primary delivery mechanism for OC1, reporting to and coordinated by the Strategic Management Committee;
- In the first three years (2026-2029), lead implementation of MG1.1 and support transition to MG1.2;
- Deliver tangible, policy-relevant outputs that can be showcased at the Triennial Cryosphere Summits;
- Coordinate with other Decade Working Groups, particularly those addressing modelling and prediction (OC2) and hazards and impacts (OC3), to ensure coherence across the Decade;
- Engage with existing international, regional, and national observation initiatives to ensure complementarity and avoid duplication, building on existing mechanisms whenever possible.

Throughout the decade, this working group will report annually to the Strategic Management Committee, on progress towards the following activities and deliverables:

1. A comprehensive inventory of cryosphere observation systems, including metadata to identify spatial, variable, and methodological gaps;
2. Facilitate the strategic enhancement of observations using emerging technologies, community-based monitoring, indigenous observations, and targeted investments;
3. Ensuring that cryosphere observations are usable for prediction, including model initialization, data assimilation, validation, and uncertainty assessment;
4. Strengthening trust, transparency, and accessibility of observational data, through interoperable standards, metadata, and including appropriate recognition of data providers;
5. Facilitating alignment between observation systems and the needs of modelling (OC2), hazards and risk assessment (OC3), and policy users.

Structure

Working Group 1 is open to all interested experts, stakeholders and relevant organizations contributing to the objectives of the Decade. Participation shall not be limited by a fixed membership cap.

To support coordination, continuity, and efficient functioning, a **Core Group of 12 members, with a Secretariat, Lead and at least one Associate Lead,** shall be established. Associate Leads work closely with the Lead to support the delivery of the Working Group's objectives and fulfil the responsibilities of the role. Members of the Core group shall be nominated by UNESCO following a call for participation, with due regard to scientific excellence, geographic balance, interdisciplinary expertise, and diversity of gender, generation, and background.

It shall guide the work of the broader Working Group, prepare meetings, facilitate follow-up on agreed actions, and report on progress to the Decade governance structure.

Participation in the Working Group is voluntary and non-remunerated. Membership of the Core group shall be reviewed every three years to allow renewal and the regular inclusion of new voices. Transitional overlap between outgoing and incoming officers may be maintained for up to six months to preserve continuity.

Modus Operandi

- The Working Group shall operate as an open expert network supported by a Core Group, which will coordinate activities, prepare work plans, monitor progress, and report to the Strategic Management Committee.
- The Working Group shall meet primarily online, with a frequency sufficient to maintain momentum, complemented by periodic in-person meetings aligned with major international events where feasible and provided funding resources are made available.
- The Working Group shall hold at least one joint meeting per year with the other Decade Working Groups (OC2–OC5) to foster synergy, share progress on cross-cutting issues, align activities across Overarching Challenges, and avoid duplication of effort.
- Progress and outputs of the Working Group shall be reported biannually by the Core group to the Strategic Management Committee and showcased at the Triennial Cryosphere Summits. Initial and subsequent appointments of Leads, Associate Leads and the Secretariat are approved by the Strategic Management Committee.
- Draft agendas and documents shall be circulated in advance of each meeting. Decisions of the Core Group and the Working Group shall be made by majority.
- Members of the Strategic Management Committee shall be expected to consult and engage with their respective networks and constituencies, and to facilitate broad participation in the activities of the Working Groups. The Lead of the Working Group serves as a member of the Strategic Management Committee and maintains regular contact with the IHP Secretariat.

Website and further details

Proceedings of the Decade, consultation summaries and deliverables will be published on the Decade's official website: www.un-cryosphere.org